

Simple LINE FOLLOWER ROBOT Using LM358

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Presented here is the design for a relay-based line follower robot without using a microcontroller (MCU). The robot provides a simple application to get started with sensor-based projects for newbie electronic hobbyists and enthusiasts. It can follow a black line/track (on white surface) or a white line/track (on black surface). The authors' prototype of the robot

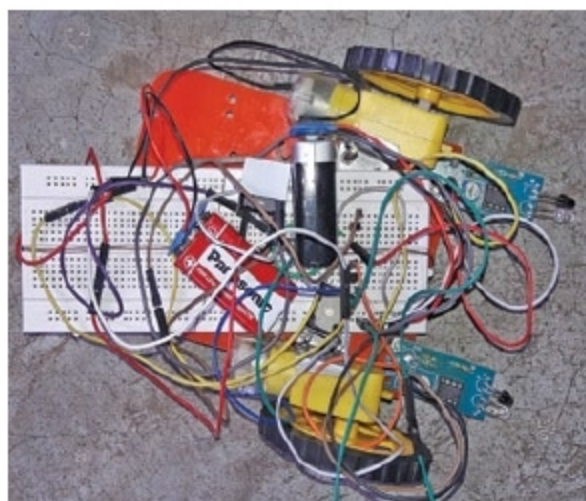


Fig. 1: Authors' prototype of the robot

is shown in Fig. 1. Block diagram of the project is shown in Fig. 2.

Circuit and working

The circuit diagram of the line follower robot is shown in Fig. 3. It is built around infrared (IR) TX LEDs (LED1 and LED3), two IR RX photodiodes (LED2 and LED4), dual-operational amplifier LM358 (IC1), two rectifier 1N4007 diodes (D1 and D2), two LEDs (LED5 and LED6), two single-changeover relays (RL1 and RL2), two

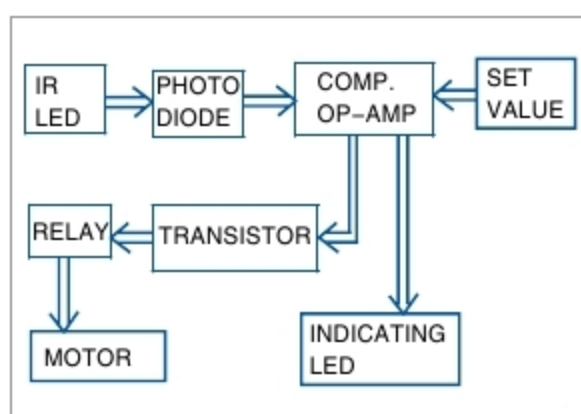


Fig. 2: Block diagram of line follower robot

10rpm geared motors and a few other components, including two wheels and a cabinet.

Sensor. Two pairs of IR transmitter (IR TX) and IR receiver (IR RX) sensors are used. When a surface is detected by IR LED, IR rays emitted by IR TX are reflected onto photodiode IR RX, which turns on the photodiode in conducting state. Signals from IR RXs are fed to non-inverting inputs of LM358.

Comparator. LM358 operational amplifier (IC1) is used as a comparator that acts as an analogue signal processor. The comparator can be considered a binary output device, since it provides only two values/states of output in volts: $+V_{cc}$ and 0V.

In this project, if V_1 and V_2 are the voltages applied to non-inverting and inverting input terminals, respectively, of LM358, then output of comparator $V_{out} = +V_{cc}$, if $V_1 > V_2$, otherwise $V_{out} = 0V$.

Transistors. Bipolar junction transistors (BJTs) are used as relay drivers. Outputs OUT1 and OUT2 of IC1 are connected to the bases of transistors T1 and T2 through resistors R5 and R8, respectively. Here, each transistor is used as a switch to drive the corresponding relay.

Fig. 4 shows the prototype module with the sensor and LM358 used in this project. V_{cc} , gnd and output terminals are shown in the module.

Relay. Pin diagram of the relay is shown in Fig. 5. When current flows through the coil of the relay from pins 5 to 1, it results in short-circuit between pins 3 and 2. However, pins 2 and 4 of the relay are shorted if no energising current is passed through the coil. A diode is

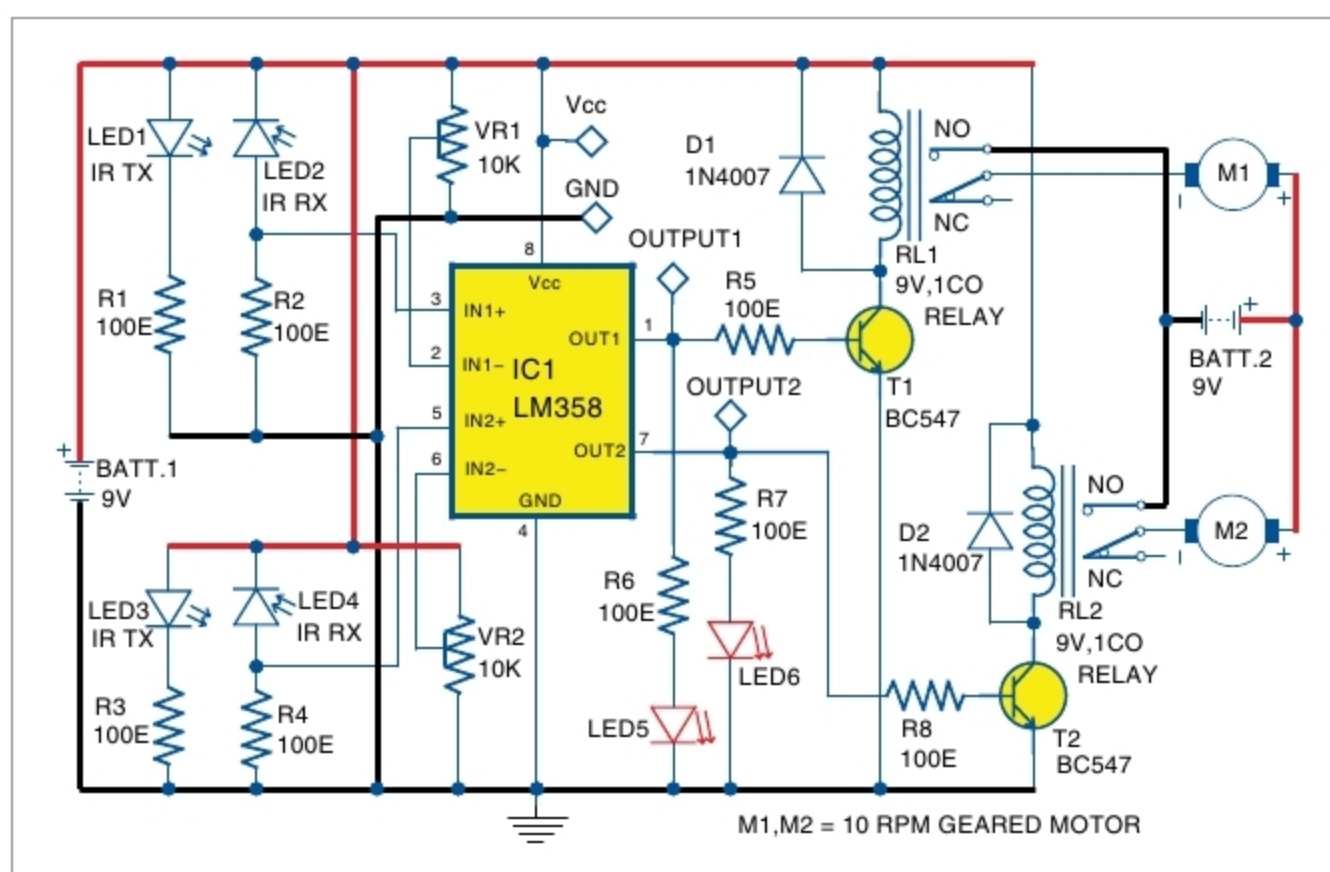


Fig. 3: Circuit diagram of line follower robot using LM358